

Measuring cell aspect ratio

Adapted from the lab compendium Snowflake Yeast Protocols — Ratcliff Lab / MuLTEE (V2).

Measuring cell aspect ratio

1. Follow the 5-day re-acclimation protocol ([[Daily growth and settling selection]] and [[Re-viving strains from frozen stock]]).
2. On the final day, transfer 100 μL of culture into fresh YPD and incubate for 12 h.
3. Stain with Calcofluor White (5 μM final), 30 min in the dark.
4. Image at $\times 40$ objective, UV excitation of the blue fluorescent cell wall stain (Nikon Ti-E system).

Image analysis pipeline (cell segmentation): zipped folder containing pipeline is attached — Cellpose-based segmentation of individual cells, measures aspect ratio (AR) directly along with area, major/minor axis, circularity, and solidity.